

# EUREKA: Phase 3 - Activity Report

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**Abstract**—During Phase3, our improvements go as follow: (1) We introduce a new validation strategy to demonstrate how during Phase2 a single-validation test is susceptible to overfitting, leading to modeling practices that do not generalize well. (2) To simplify researchers use of this technology we abstract modeling behind a dedicated framework called ADRDModel, this allows identical solutions to be executed over diverse set of studies, with no changes required, further demonstrating generalization. (3) Six different studies are explored, including original Phase2 data, extended MHAS, another four international studies, and access to restricted data from NACC. (4) Performance improvements are evaluated over 15 different experiments, ranging from automatic feature selection, to more advanced process like biometrics and leveraging pre-diagnosis MRI/PET scans when available. (5) Automatic bias mitigation is introduced for ADRDModel, giving researchers the ability to define cohorts of interest which will be automatically optimized. (6) Explainability has improved with a built-in uncertainty estimation as well as per-patient diagrams of the decision process. (7) We introduce the concept of HealthScore, a patient-aware probabilistic model to determine likelihood of strong cognitive decline, this is in order to compensate the natural variance that exist between individuals even, for those with the same cognitive-score.

## I. PHASE 2 - SUMMARY

During Phase2, our model simplifies the multi-year nature of the dataset by duplicating records and allowing the model to determine the appropriate usage of *year* signal. For most of our modeling we relied on LightGBM [4] and CatBoost [5] as they are state-of-the-art for tabular data [2], as well as showing great results for Alzheimer’s Disease Prediction [3].

Parameter optimization was kept at minimal to prevent overfitting as we observed fluctuations up to  $\pm 1.28$  RMSE standard deviation (SD), enough to group Top #6 competitors in the Social determinant track. We helped to control overfitting by tuning *path\_smooth*. Feature engineering provided the best gains, and Target-based encoding for both numerical and categorical features gave up to 1.48 leaderboard improvement.

Modeling wise, we introduce the concept of *decline-acceleration*, a way to estimate not only *what* is the expected next cognitive score, but with how much acceleration the decline has occurred with respect to previous year, this was key for a Top #3 final leaderboard position.

In the following sections we will discuss how our model has evolved as part of Phase3 improvements.

## II. MODEL ARCHITECTURE DESIGN

Although our Phase2 final solution includes three models: main LightGBM, secondary CatBoost and, third, acceleration-model, we opted to remove both secondary and acceleration

model for the sake of simplification in Phase3. This provides an easier explainable system, reduces dependencies and solution complexity.

Target-based encoding proved useful during Phase2 evaluation but increases the risk of overfitting over multiple studies, this was removed in our Phase3 model and instead all categorical features are handled using the ordered-target-based encoding built-in directly in LightGBM. This eliminates all pre-processing in Phase3, which has none, a great advantage over Neural Network based approaches.

A new modality for uncertainty estimation was added in Phase3, based on K-fold partition, each model provides slight variants that can be used for uncertainty, this is abstracted and automatically handled by our Alzheimer’s Disease and Related Dementias Model (ADRDModel) package.

In order to prove the consistency of ADRDModel, its parameters are fixed to allow multi-study high-performance without relying on expensive re-optimizations, to prevent overfitting, *num\_leaves* has been reduced from 32 to only 8, as further increases come with marginal metric boost that does not always translate to better performance in other studies beyond Mexican Health and Aging Study (MHAS). In order to enhance explainability, both feature-fraction and bagging-fraction are further reduced from 0.6 (in Phase2) to 0.4 in Phase3 which forces the model to identify confounding factors, beyond just raw loss reduction.

In Figure 1, we show both Phase 2 & 3 solution architectures.

The Feature Fetch module provides an abstraction layer that has been optimize with feature-selection mechanism of pre-existing good features. This allows very fast integration between datasets, especially very large ones like The National Alzheimer’s Coordinating Center (NACC) Uniform Data Set (UDS), during this module execution all known available useful features are extracted and engineered, this includes social aspects, but also other resources when available, like biometrics, historical previous cognition tests, and Magnetic Resonance Imaging (MRI) & Positron Emission Tomography (PET) scans.

Since ADRDModel has been generalized to operate over multiple studies, with no changes required, we instead defined a normalized study-agnostic dataset; this process is completed by the Dataset Adapter, which takes a study-specific data, like MHAS or NACC-UDS and harmonize it further following a common standard, this allow then an identical ADRDModel to be executed for any dataset, regardless of the study origin.

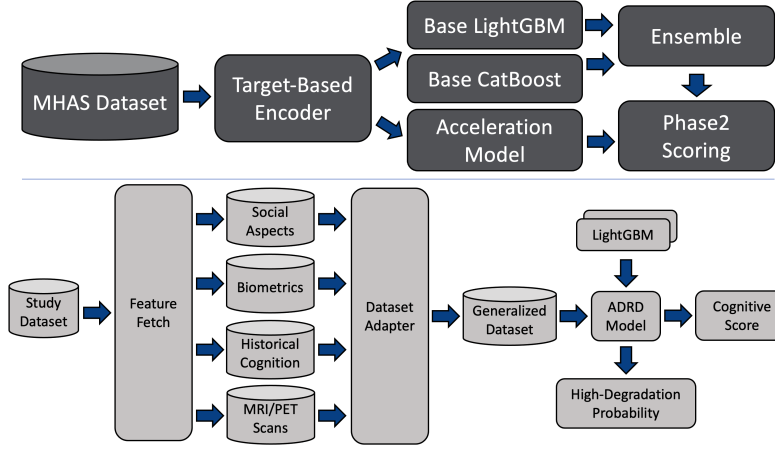


Fig. 1. Phase-2 (top) vs Phase-3 (bottom) Model Architecture Design

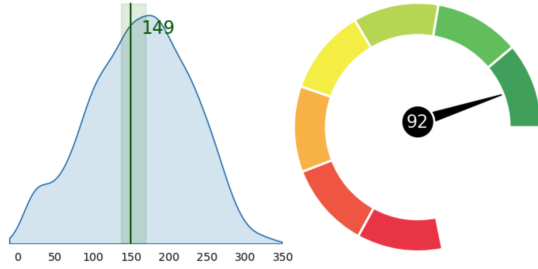


Fig. 2. Cognition Strenght & Health Score ADRDModel Output

Our ADRDModel has then the capability to provide a multi-output score, both indicating the expected cognitive score over a future-date (normally 4 year mark) as well as Health Score, a probability based estimate of how likely this specific patient might experience severe degradation in the future (higher is better), this is important as people will have inherently different cognition score due to natural decline (e.g. aging) yet it might be a reasonable level for their particular group (e.g. age-wise), the outputs of the model can be seen in Figure 2.

### III. DATA COLLECTION & PREPARATION

To ensure our design is adaptable to a variety of scenarios, we collected multiple studies datasets as shown in Table I.

There is a small observed difference between Phase2 data and our pre-process version from MHAS, but with less of 1% patient difference there is no significant impact. Different available features can be obtained from our Feature Fetch module, if *FeatureSelection* is activated, when using the entire feature space from MHAS, over 3000 features are available, yet, highest-performance is possible with only 14% of feature space, although this is almost 2x the features available from Phase2.

The HRS dataset provides great historical tracking of waves to be able to evaluate different time-horizon prediction mechanisms, and comes with the largest set of features. Although based on our FeatureFetch module, we identified around 15%

TABLE I  
ADRDMODEL SUPPORTED DATASETS

| Dataset                                       | Reference   | # Samples | # Features   |
|-----------------------------------------------|-------------|-----------|--------------|
| DrivenData Eureka Challenge                   | Eureka [6]  | 2,002     | 183          |
| Mexican Health & Aging Study                  | MHAS [7]    | 1,983     | 474 to 3384  |
| United States Health and Retirement Study     | HRS [8]     | 4,181     | 1090 to 1681 |
| Longitudinal Aging Study in India             | LASI [9]    | 35,363    | 461          |
| Costa Rican Longevity and Healthy Aging Study | CRELES [10] | 2,361     | 142          |
| National Alzheimer's Coordinating Center      | NACC [11]   | 8,991     | 305          |

of features are responsible for most of the value, this is managed automatically for medical researchers.

The LASI dataset is by far the largest, with over 30,000 samples, even after accounting for only single-patient per-household. As part of our DatasetAdapter module we provide automatic patient selection with the *subsample* parameter, this provides easy exploratory analysis.

The CRELES dataset provides multiple waves, but most signals are missing, the *FeatureFetch* module automatically adjust usable modeling signals, ensuring no time-leakage as target preparation is always estimated using future waves.

The NACC is a restricted access dataset and cognitive scores are not immediately available, however, to enhance this study, our *FeatureFetch* module can be linked with the Alzheimer's Disease Sequencing Project (ADSP), and automatically link results from the Phenotype Harmonization Consortium (PHC) dataset [12] funded by NIA (U24 AG074855, U01 AG068057 and R01 AG059716). We also extended this with access to historical medical records including MRI and PET scans.

All datasets are further harmonized to be processed by the same ADRDModel. The High-Degradation Probability output, can be evaluated using AUC metric which can be compared across different studies. Raw Cognitive scores do not provide this advantage, since they are scale dependent, to aid this,

TABLE II  
PHASE3 EXPERIMENTS SUMMARY

| Dataset | Experiment                                                  | RMSE                   | AUC                   |
|---------|-------------------------------------------------------------|------------------------|-----------------------|
| Eureka  | Phase2 Model                                                | 40.657<br>$\pm 1.688$  | 0.8807<br>$\pm 0.023$ |
| MHAS    | Phase2 Equivalent<br>Extended Features                      | 38.344<br>$\pm 0.468$  | 0.906<br>$\pm 0.019$  |
| MHAS    | With Biometrics                                             | 38.538<br>$\pm 1.46$   | 0.9059<br>$\pm 0.02$  |
| MHAS    | Biometrics & Historical Cognition                           | 33.6134<br>$\pm 2.005$ | 0.9282<br>$\pm 0.022$ |
| MHAS    | Biometrics & Historical Cognition<br>with Feature Selection | 33.405<br>$\pm 2.071$  | 0.9285<br>$\pm 0.022$ |
| HRS     | Missing features removed                                    | 0.8454<br>$\pm 0.025$  | 0.7704<br>$\pm 0.019$ |
| HRS     | With Feature Selection                                      | 0.8438<br>$\pm 0.024$  | 0.7709<br>$\pm 0.019$ |
| HRS     | With Feature Selection<br>& Education Bias Mitigation       | 0.8434<br>$\pm 0.023$  | 0.7732<br>$\pm 0.018$ |
| MHAS    | Transfer learning<br>from HRS Model                         | 33.579<br>$\pm 2.177$  | 0.9286<br>$\pm 0.023$ |
| HRS     | Transfer Learning<br>From MHAS Model                        | 0.8446<br>$\pm 0.025$  | 0.7717<br>$\pm 0.02$  |
| LASI    | With Feature Selection                                      | 0.6104<br>$\pm 0.006$  | 0.9066<br>$\pm 0.003$ |
| CRELES  | With Feature Selection                                      | 0.8346<br>$\pm 0.055$  | 0.8096<br>$\pm 0.011$ |
| NACC    | With Feature Selection                                      | 0.3797<br>$\pm 0.008$  | 0.7546<br>$\pm 0.014$ |
| NACC    | MRI Scan Data Only                                          | 0.6667<br>$\pm 0.013$  | 0.872<br>$\pm 0.015$  |
| NACC    | PET Scan Data Only                                          | 0.665<br>$\pm 0.025$   | 0.927<br>$\pm 0.019$  |

we further introduce automatic Z-normalization as part of the *DatasetAdapter* module.

#### IV. EXPERIMENTS & RESULTS

Our first improvement comes from changing the validation mechanism and introducing new metrics and tests that are study-agnostic. As discussed in Section I, Phase2 dataset using a single-validation set can be prone to overfitting, as shown with experiments, a SD of  $\pm 1.28$  shows no real statistical significant difference between Top #6 competitors, which go from 38.203 to 38.847 RMSE, moreover, this ability to overfit leads to inaccurate representation of performance.

Our built-in k-fold validation mechanism highlights any potential overfitting pattern, to ensure improvement is real. In general, we claim an improvement when this exceeds the natural standard deviation of the multiple k-runs. All experiments are reported in Table II, RMSE is shown for the continuous scores (cognitive strength), and AUC for the Health Score, note that for datasets without z-normalization as was the case in Phase2 data, RMSE is not comparable with other studies, but AUC is generic across them.

From the results we can observe the importance of feature selection, which is done automatically by our FeatureFetch module, in experiments where we deliberately turned it on/off to study the impact, we constantly observe improvement, not only in performance, but in model simplicity and explainability. For MHAS we can reduce as much as 85% of features, this aligns with the Phase2 data from Eureka

Challenge, which uses much fewer features to still provide reasonable performance. HRS shows similar pattern of improvement. Note not all experiments are comparable, such as MRI and PET scans where data is so sparsely available, only patients with known historical, pre-diagnosis studies are included, yet it provides insights on the value of these resources.

More advanced experiments, like Bias Mitigation will be discussed in Section V. Similarly, our approach to explore transfer learning between studies is reviewed in detail in Section VI

#### V. GENERALIZATION, BIAS & EXPLAINABILITY

With the development of ADRDModel algorithm, we hope to close the gap between ML and medical practitioners, a single model is capable to provide high-performance result without any modification for 6 different studies and over 15 variations of features and data-sources, all without any ML knowledge required. With the DatasetAdapter module, we further allow expansion and enable researchers to focus on data alone, with a data ingestion protocol one can easily expand ADRDModel to new studies without requiring any modification to its internal structure.

Bias mitigation is handled automatically and it can be adjusted by researchers with a parameter in ADRDModel called *bias\_cols*, this will proactively adjust the losses by overweighting groups poorly represented, in Table II we show how education can be used to force the model to focus equally to all segments, bringing overall improvements, and reducing performance gaps for each group.

Improvements in explainability are made directly in ADRDModel, for each prediction, researchers can access both cognitive strength estimation, as well as our health-score probabilistic output. Top variables responsible for decision outcomes are plot including their contribution to the final score. Since ADRDModel is natural built with uncertainty estimation, ranges are provided for each model, as well as final prediction as shown in Fig 3.

#### VI. CROSS-STUDY SELF ADAPTIVE PRE-TRAINING

One innovation introduced is the study-agnostic-pretrain, a review of how the study in Mexico with MHAS can provide value to the HRS in USA, and vice versa. Preliminary results are promising but still low impact. When using HRS learnings to improve MHAS, we noticed a flat response in performance. The new MHAS model does leverage patterns learn from HRS, but it uses them as replacement, and not additions to the patterns found in MHAS, as result we obtain a more compact model (e.g. fewer features required), but similar performance. When evaluating the opposite direction, MHAS as an improvement to HRS, a similar pattern showed up, with improvements like HealthScore, but below standard deviation, and as a result tagged as statistical insignificant.

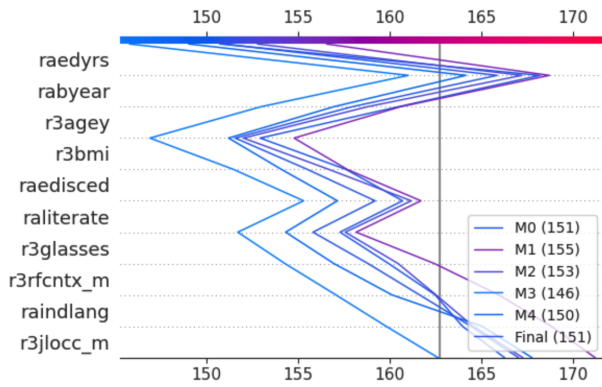


Fig. 3. ADRDModel Explainable Decisions: MHAS Example

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